Sampling in Research – PDF Handout

1. What Is Sampling?

Sampling is the process of selecting a smaller group from a larger population so researchers can study the sample and generalize the findings to the population. It makes research practical, efficient, and cost■effective.

2. Why We Sample

- Populations are often too large to study fully.
- Sampling reduces cost and time.
- Some groups are hard to reach.
- Allows research without destructive testing.
- Produces reliable results when done properly.

3. Key Concepts

- Population: Entire group of interest.
- Sample: Subset of the population.
- Sampling Frame: Listing of population members.
- Sampling Unit: Single element selected.
- Sampling Design: Plan for selecting the sample.

4. Steps in Sampling

- Define population.
- Identify sampling frame.
- Select sampling technique.
- Determine sample size.
- Apply procedure consistently.

5. Probability Sampling

- Simple Random: Equal chance for each member.

- Systematic: Select every kth element.
- Stratified: Sample from population subgroups.
- Proportional Stratified: Sample size matches subgroup size.
- Cluster: Select whole clusters when populations are widespread.

6. Non■Probability Sampling

- Convenience: Based on availability.
- Purposive: Based on expertise or relevance.
- Quota: Fill specific subgroup quotas without randomness.

7. Sampling Bias

Bias occurs when selection methods produce an unrepresentative sample. It harms accuracy and generalization.

8. Reducing Bias

- Use random methods.
- Maintain accurate sampling frames.
- Train data collectors.
- Avoid convenience■only selection.

9. Practical Examples

University study habits: random, stratified, or convenience sampling.

City electricity survey: cluster or systematic sampling.

10. Conclusion

Proper sampling allows reliable generalization, saves resources, and strengthens research quality.